

Firefly estimator, the math

What runs in `src/brain` as of Sun Sept 13, before the method freeze. Constants are the ones in code.

The problem, in the Defense brief's own form

$$y = H x + a + \varepsilon$$

x is the true temperature of every space. H selects the spaces that have a sensor. ε is ordinary noise. a is the corruption vector: zero for honest sensors, arbitrary for at most k of them. The estimator is not told which entries of a are nonzero. The Kalman baseline assumes $a = 0$. Ours does not.

1. Shared physics model (both brains)

$$x_i(t+1) = x_i(t) + \sum_j r_{ij} (x_j(t) - x_i(t)) + c (T_{\text{amb}} - x_i(t)) + b_i \cdot g (T_{\text{flame}} - x_i(t))$$

r_{ij} is the per-edge heat-transfer rate from the structure plan (door 0.15, passage 0.2, bulkhead 0.05, and so on). $c = 0.02$ is loss to ambient. $T_{\text{flame}} = 900$ °C, $g = 0.25$ (1.5× in a fuel-hazard space). b_i is 1 if space i is burning. The first term is a weighted graph Laplacian, so the model is $x(t+1) = A x(t) +$ generation with A built from the plan's edges. The structure is data; the estimator is not.

2. Kalman baseline

```
predict:   $\hat{x} \leftarrow A \hat{x} + 40 \cdot b$        $P \leftarrow A P A^T + Q$        $Q = 25 \cdot I$ ,  $R = 4 \cdot I$ ,  $P_0 = 100 \cdot I$ 
update:   standard Kalman gain over every reading, none doubted
 $b_i = 1$   if  $\hat{x}_i > 200$ 
confidence = mean_i (  $1 - \sqrt{P_{ii}} / 50$  )
```

It knows the building's physics exactly. Its confidence comes from its own covariance P , which only shrinks as readings arrive. It never asks whether the readings agree with each other, so once converged it reports about 0.97 whether or not it is right.

3. Ours, step one: decide which readings to trust

Each reading is tested against physics and its neighbors before it is used. A reading is set aside as *suspect* if it is:

stale	timestamp lags the current tick by more than 2
frozen	unchanged ($ \Delta < 0.05$ °C) for 6 ticks while the median neighbor moved > 15 °C
impossible	rose faster than the model allows from last tick's estimate
cold	below ambient + 10 °C while trusted neighbors read above 250 °C
pathless	above 200 °C with nothing within two edges above 60 °C (3-tick grace at the start)

Readings that pass form the trusted set y_T . Suspect readings are not used for scoring and are reported to the operator.

4. Ours, step two: score hypotheses, tolerating k liars

A hypothesis **B** is a set of burning spaces. Each candidate is rolled forward 3 ticks from the estimate 3 ticks ago with the model in section 1, giving predicted temperatures $\hat{x}_B$. Its score is the mean absolute residual over trusted readings after discarding up to k of the largest *downward* misses:

$$s(B) = \text{mean}_{\{i \in T\}} | \hat{x}_{B,i} - y_i | \quad \text{dropping up to } k \text{ terms where } \hat{x}_{B,i} > y_i \quad (k = 1 \text{ by default})$$

Only downward misses can be dropped, because every failure mode in the model (freeze, blind, saturate, flashover) pushes a reading down or away from the truth, never up. A sensor reading *hotter* than a hypothesis allows is never excusable, so the "no fire" hypothesis cannot discard the one sensor that sees the fire. This is the sparse-corruption tolerance from the brief: the estimate may disagree with k sensors without knowing which k .

5. Ours, step three: keep the set of survivors, not one winner

```
s* = min_B s(B)
kept = { B : s(B) ≤ s* + max(8 °C, 0.15 · s*) }
burning = n kept           spaces in every surviving hypothesis
ambiguous = u kept - n kept spaces in some survivors but not all
```

Ties break toward the hypothesis that explains the data with no liars at all. A space whose *trusted* reading has been above 250 °C for 3 consecutive ticks is forced into every candidate, for as long as its assumed fuel lasts.

6. Graded belief

Survivors are weighted like a Boltzmann distribution, with the tolerance as the temperature, so a hypothesis at the edge of tolerance weighs e^{-1} of the best:

$$w_B = \exp(-(s(B) - s^*) / \tau) \quad \tau = \max(8, 0.15 \cdot s^*)$$
$$P(\text{space } i \text{ burning}) = (\sum_{\{B \ni i\}} w_B) / (\sum_B w_B) \times 0.8^{(\text{suspect sensors in space } i)}$$

7. Earned confidence

```
conf = ( 1 / |kept| ) × 0.8^(#suspect sensors) × min(1, 6 / s*)
capped at 0.7 until the burning set is uncontested and unchanged for 5 consecutive ticks
floor 0.05, and exactly 0.05 when no reading at all is trusted
```

Confidence is one over the number of surviving explanations, discounted for every sensor being ignored and for poor fit, and held below 0.7 until the answer has stood still. This is why ours is never confidently wrong, and also why it takes 30 to 56 ticks to reach high confidence in the sweep: the caution is the price.

The slide version

$$y = H x + a + \varepsilon$$

Kalman: assume $a = 0$
average everything

Ours: $s(B) = \text{mean } |\hat{x}_B - y|$
drop $\leq k$ downward misses
report $\{ B : s(B) \leq s^* + \tau \}$

Same physics, same readings, same seeds. The difference is whether readings are allowed to lie, and whether the answer is a point or a set.

Reported metrics (from computeMetrics)

false certainty	fraction of ticks with $\text{conf} \geq 0.9$ and burning set $\neq$ truth
estimation error	mean $ \text{estimate} - \text{truth} $ over spaces, ticks $\geq$ onset
time to recovery	ticks from onset to the first of 5 consecutive exact matches
compute cost	wall-clock ms per brain.step
wrong dispatch	fraction of ticks naming a space that is neither burning nor ambiguous
ambiguity coverage	fraction of ticks where $\text{truth} \subseteq \text{burning} \cup \text{ambiguous}$
Brier score	mean over spaces and ticks of $(P(\text{burning}) - \text{truth})^2$